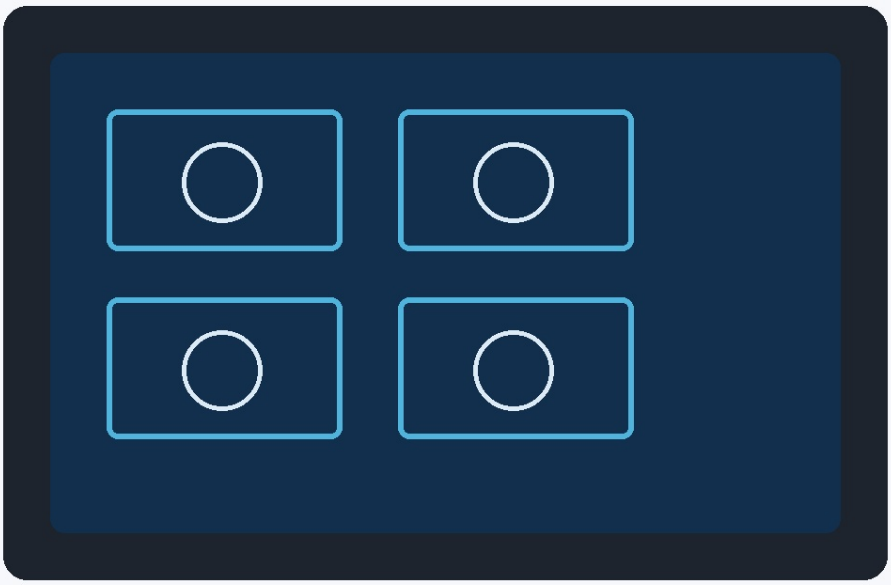
NEXASTORE

AI Machines • Developer Workstations • Next-Generation Computing



VISION + DL

Hardware for Computer Vision and Deep Learning

Plan GPU, VRAM, RAM and storage for visual AI

NexaStore Knowledge Series • 2026 Edition

1. Visual AI Has Different Bottlenecks

Computer vision workloads combine image decoding, augmentation, tensor compute, model memory and dataset I/O. Training generally demands more GPU memory than inference, while high-resolution images and larger batch sizes can increase memory use sharply.

Workload	GPU direction	RAM	Storage direction
Image classification learning	8 GB VRAM useful	32 GB	1 TB NVMe
Object detection development	12-16 GB VRAM	32-64 GB	1-2 TB NVMe
Segmentation / larger images	16-24+ GB VRAM	64 GB	2 TB+ NVMe
Fine-tuning vision models	16-24+ GB VRAM	64-128 GB	2-4 TB NVMe
Large training jobs	High-VRAM / multi-GPU cloud	128 GB+	Fast multi-TB storage

2. Why VRAM Usage Grows

Factor	Impact
Image resolution	Larger tensors consume more accelerator memory.
Batch size	More samples processed together increase activation memory.
Model architecture	Larger backbones and transformers can require substantially more memory.
Training	Gradients, activations and optimizer state make training heavier than inference.
Precision	Lower-precision modes can reduce memory and improve throughput when supported.

3. CPU, RAM and Storage Still Matter

Resource	Role
CPU	Image decode, augmentation, data loading, preprocessing and orchestration.
System RAM	Caches datasets, supports data loaders, notebooks, IDEs and background services.
NVMe SSD	Reduces data-loading bottlenecks for large image/video datasets.
GPU	Accelerates tensor operations for training and inference.
Network	Important when datasets/models reside in cloud or network storage.

4. Example Machine Profiles

Profile	Suggested direction	Best for
CV developer laptop	32-64 GB RAM, 12-16 GB VRAM	Prototyping, inference, moderate training
AI workstation	64-128 GB RAM, 24-32 GB VRAM	Frequent training and larger experiments
Cloud-first developer	32 GB RAM, GPU optional	Managed GPU training with local coding/data prep
AI lab	128 GB+ RAM, high-VRAM GPU	Large datasets, multiple models and heavier experimentation

5. Recommendation Questions

• Are you training models locally or only running inference?

• What image/video resolution and batch size do you expect?

• Does the framework require a specific GPU software ecosystem?

• How large is the dataset and where will it be stored?

• Can heavy training move to cloud GPU while development stays local?

6. NexaStore Agent Rule

A recommendation agent should use structured catalog data for exact GPU, VRAM, RAM, storage, price and availability, then use this guide to reason about suitability for classification, detection, segmentation or training.

Educational notice

This fictional guide provides approximate planning ranges, not compatibility guarantees.